

Low level of catalase plays a major role in greying process of human hair.

4.10.2 Structure of Enzymes

Enzymes are generally globular proteins that range in molecular weight from about 10,000 to several million. Each enzyme possesses a unique sequence of amino acid that causes it to fold into a characteristic three dimensional shape with a specific surface configuration. This enables it to find the correct substrate from among the large number of diverse molecules in the cell.

A molecule acted upon by an enzyme is called a substrate. Enzymes are specific and act on specific substrates and each enzyme catalyzes only one reaction. Enzyme consists of a protein portion, named apoenzyme and a non protein component, named cofactor (Figure 4.11).

Apoenzyme (Protein portion) (Inactive form)	+	Cofactor (Non protein) (Inactive form)	⟶	Holoenzyme (Active enzyme)

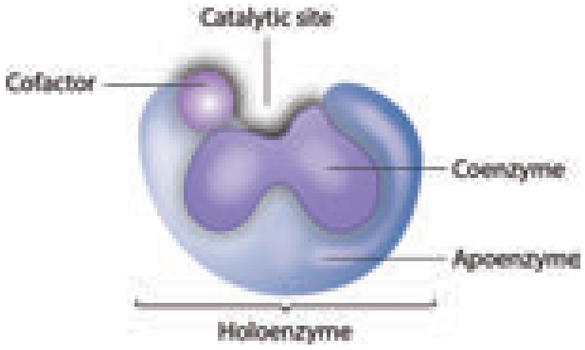


Figure 4.11: Structure of enzyme

The region of an enzyme where substrate molecules bind and undergo a chemical reaction is its active site. Each active site is specially designed in response to their substrate; as a result most enzymes have specificity and can only react with particular substances. After the formation of enzyme substrate complex (Figure 4.12), forces exerted on the substrate by the enzyme cause it to react and become the product of the intended reaction.

Example: Sucrase catalyses the hydrolysis of sucrose to glucose and fructose.

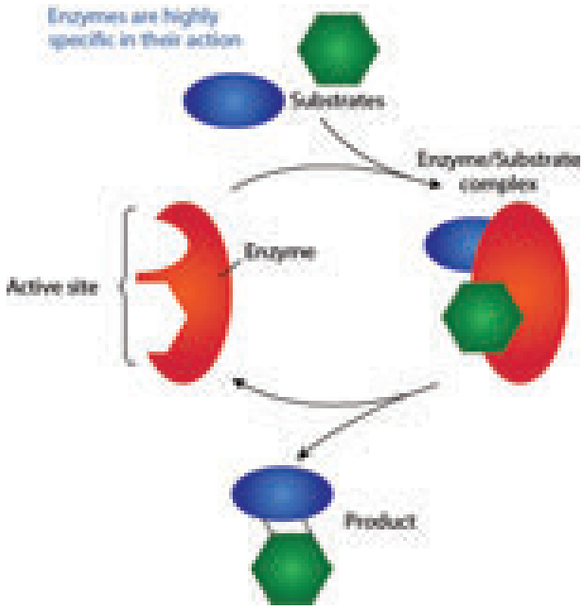


Figure 4.12: Mechanism of product formation from substrate through Enzyme

Apoenzyme is the inactive form of the enzyme which gets activated after binding with a cofactor. Coenzymes are small organic molecules that can be loosely or bound to an apoenzyme and they transport chemical group from one enzyme to another.

Cofactor is a chemical compound or metallic ion that is required for enzyme activity. Example: NAD^+ is derived from

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